



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI BYRAVESHWARA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

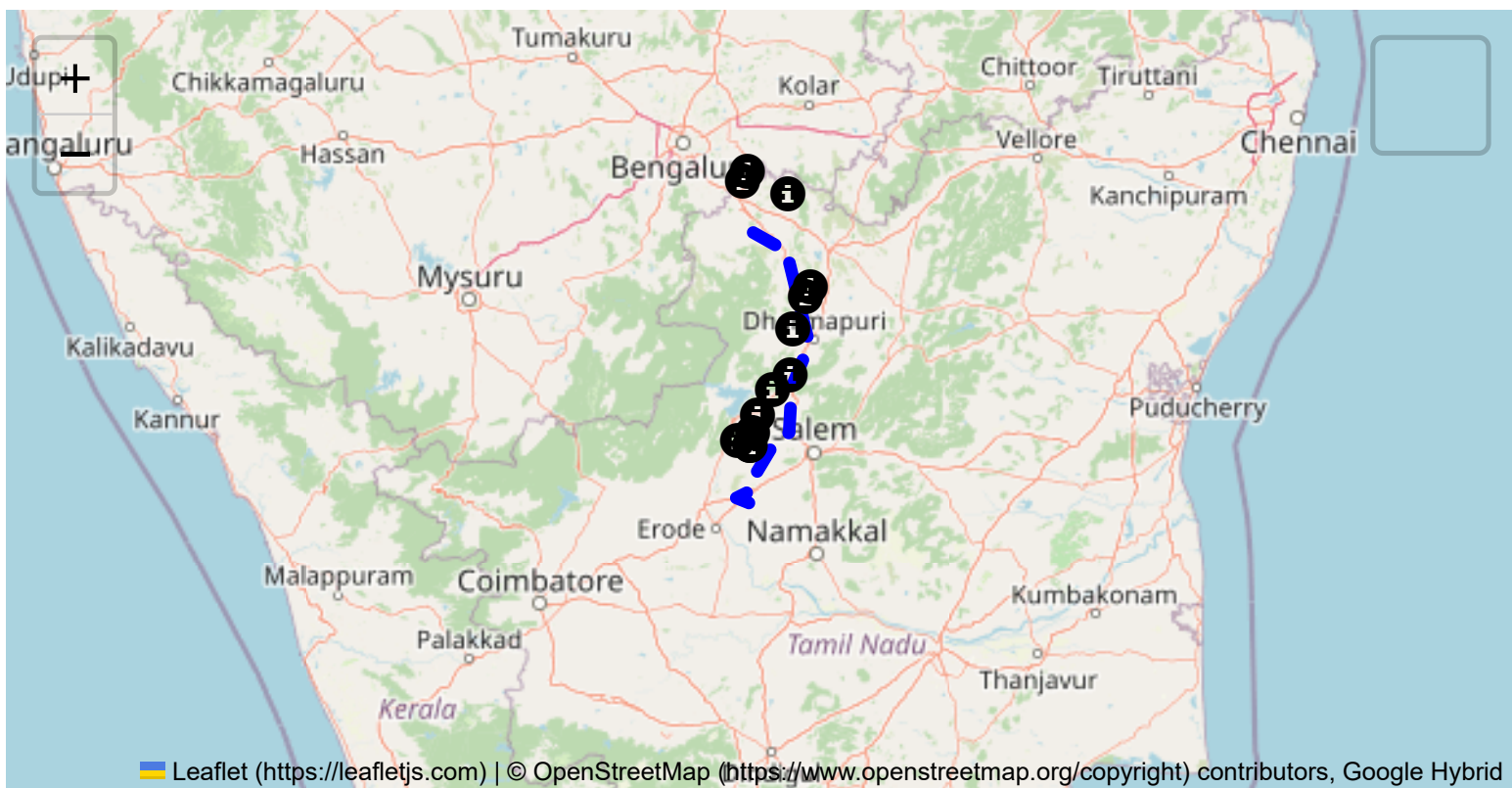
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

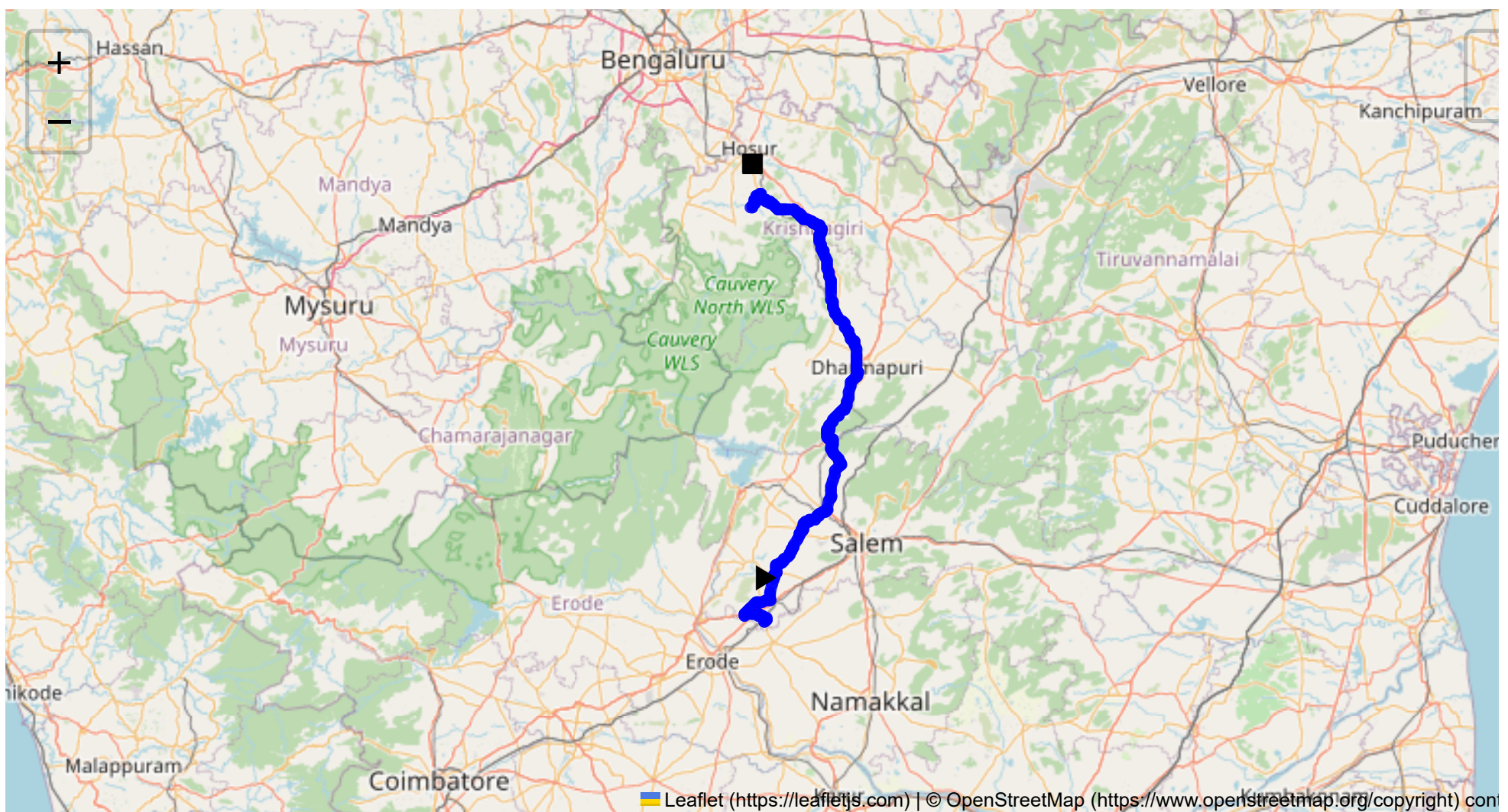
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 178.38 km
Estimated Duration: 3.7 hours
Adjusted Duration (Heavy Vehicle): 4.6 hours
Start: (11.4381, 77.8734)
End: (12.568641, 77.832935)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Kouthalam covers approximately 178.38 kilometers in Tamil Nadu, India, passing through various towns and highways. It includes stops or key points at Puthur, Padaiveedu, Sangagiri, Erumaipatti, Desavilakku, Omalur, Thoppur, Dharmapuri, Virupakshipuram, Rayakottai, and Uddanapalli. The journey takes roughly 3.68 hours for heavy vehicles, relying on state highways and main roads with varying traffic conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu experiences a tropical climate. The journey may encounter hot and humid weather, especially between March and May. The monsoon season from June to September can bring heavy rainfall, leading to waterlogging and reduced visibility. During this season, the risk of slippery roads and minor flooding is heightened.

3. Analysis of Traffic Patterns: Traffic congestion is typically heavier in urban areas such as Omalur and Dharmapuri, with peak hours generally between 8-10 AM and 5-8 PM. Rural parts of the route might see less traffic but watch for slower-moving agricultural vehicles and occasional livestock on roads.

4. Assessment of Road Quality and Infrastructure: The roads vary in quality, with main highways often well-paved and maintained. Rural roads may have potholes or less consistent surfaces, which could pose risks to heavy vehicles. Some areas might lack proper signage or have challenging turns.

5. Suggestions for Alternative Routes for Emergencies: In case of roadblocks or emergencies, alternatives include:

- Using NH-44 through Madurai if southern detours are needed.
- Local roads around Salem and Dharmapuri can offer passage but may not be suitable for heavy vehicles at all points.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Tamil Nadu requires transport permits for hazardous materials, including adherence to guidelines on packaging, handling, and safety measures. Checkpoints may verify these documents.

7. Overview of Historical Incidents: There have been occasional accidents involving heavy vehicles around Dharmapuri and Salem, often linked to driver fatigue or inclement weather. Few involve hazardous spills, but the risk remains due to winding roads.

8. Environmental Considerations and Sensitive Areas: The route passes by several agricultural zones and near water bodies. Care should be taken to prevent any spillage of hazardous materials, especially near these ecologically sensitive areas.

9. Analysis of Communication Coverage: Most of the route should have good mobile coverage, though rural sections and certain hilly areas might experience brief dead zones. Drivers should carry offline maps and emergency contact information.

10. Estimated Emergency Response Times:

- Urban areas like Salem and Dharmapuri: Approximately 15-30 minutes.
- Rural segments: Response may take up to 60 minutes due to distance from major towns.

12. Overall Summary of Risk Assessment: The route is generally safe but has challenges related to weather, road quality, and traffic patterns, especially for heavy vehicles. Adequate preparation for adverse weather conditions, familiarization with the route, and awareness of local traffic laws are critical to minimizing risk. Emergency response is reasonably efficient in urban areas but slower in remote regions. Compliance with regulations for transporting hazardous materials is necessary to ensure safe transit without incident.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.60628, 77.86014	15 KM/Hr
1	Roundabout	High	12.60637, 77.86022	15 KM/Hr
2	Turn	High	11.43961, 77.87341	15 KM/Hr
3	Turn	High	11.44077, 77.87533	15 KM/Hr
4	Turn	High	11.44898, 77.87410	15 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr
6	Turn	Medium	11.45764, 77.85219	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45545, 77.81812	30 KM/Hr
9	Turn	High	11.45584, 77.81695	15 KM/Hr
10	Turn	Medium	11.45656, 77.81709	30 KM/Hr
11	Turn	High	11.49383, 77.88442	15 KM/Hr
12	Turn	Medium	11.56352, 77.90078	30 KM/Hr
13	Turn	Medium	11.59189, 77.90645	30 KM/Hr
14	Turn	Medium	11.59328, 77.90809	30 KM/Hr
15	Turn	Medium	11.63867, 77.94841	30 KM/Hr
16	Turn	High	11.64217, 77.95200	15 KM/Hr
17	Turn	Medium	11.64712, 77.95118	30 KM/Hr
18	Turn	Medium	11.67858, 77.96834	30 KM/Hr
19	Turn	Medium	11.69776, 77.97983	30 KM/Hr
20	Turn	High	11.70233, 77.97869	15 KM/Hr
21	Turn	Medium	11.71394, 77.99572	30 KM/Hr
22	Turn	Medium	11.71916, 78.01117	30 KM/Hr
23	Turn	Medium	11.72504, 78.01680	30 KM/Hr
24	Turn	Medium	11.73766, 78.03445	30 KM/Hr
25	Turn	High	11.73651, 78.03868	15 KM/Hr
26	Turn	Medium	11.74120, 78.04588	30 KM/Hr
27	Turn	Medium	11.77998, 78.05786	30 KM/Hr
28	Turn	Medium	11.86915, 78.08382	30 KM/Hr
29	Turn	Medium	11.87998, 78.08170	30 KM/Hr
30	Turn	Medium	11.91266, 78.05852	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
31	Turn	Medium	11.92653, 78.06517	30 KM/Hr
32	Turn	Medium	11.93196, 78.06473	30 KM/Hr
33	Turn	Medium	11.93473, 78.06268	30 KM/Hr
34	Turn	Medium	11.93735, 78.05344	30 KM/Hr
35	Turn	Medium	11.95498, 78.04689	30 KM/Hr
36	Turn	High	11.95847, 78.04989	15 KM/Hr
37	Turn	High	11.96011, 78.04944	15 KM/Hr
38	Turn	Medium	11.96251, 78.05104	30 KM/Hr
39	Turn	High	11.96427, 78.05118	15 KM/Hr
40	Turn	High	11.96526, 78.04950	15 KM/Hr
41	Turn	High	11.96637, 78.04923	15 KM/Hr
42	Turn	Medium	11.96859, 78.05102	30 KM/Hr
43	Turn	Medium	12.12010, 78.13282	30 KM/Hr
44	Blind Spot	Blind Spot	12.51381, 78.03208	10 KM/Hr
45	Turn	Medium	12.51395, 78.02897	30 KM/Hr
46	Turn	High	12.51632, 78.02571	15 KM/Hr
47	Turn	Medium	12.52154, 78.02425	30 KM/Hr
48	Turn	Medium	12.52697, 78.00960	30 KM/Hr
49	Turn	Medium	12.52856, 78.00884	30 KM/Hr
50	Turn	High	12.55122, 77.97229	15 KM/Hr
51	Turn	High	12.55312, 77.96538	15 KM/Hr
52	Turn	Medium	12.56474, 77.95413	30 KM/Hr
53	Turn	Medium	12.57394, 77.89805	30 KM/Hr
54	Turn	Medium	12.58285, 77.89311	30 KM/Hr
55	Turn	Medium	12.58407, 77.88826	30 KM/Hr
56	Turn	Medium	12.58363, 77.88722	30 KM/Hr
57	Turn	Medium	12.58534, 77.88126	30 KM/Hr
58	Turn	Medium	12.58929, 77.87842	30 KM/Hr
59	Turn	Medium	12.59188, 77.86980	30 KM/Hr
60	Turn	Medium	12.59963, 77.86303	30 KM/Hr
61	Turn	Medium	12.60200, 77.86284	30 KM/Hr
62	Turn	High	12.60468, 77.85443	15 KM/Hr
63	Turn	Medium	12.60203, 77.85299	30 KM/Hr
64	Turn	Medium	12.59337, 77.84801	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
13	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
14	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
15	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
20	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
22	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
24	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
27	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium
29	clinic	Kowthalam HSC	12.5676496, 77.8312573	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
21	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
23	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.60628, 77.86014



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.60637, 77.86022



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44077, 77.87533



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45545, 77.81812



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45584, 77.81695



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45656, 77.81709



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56352, 77.90078



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59328, 77.90809



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69776, 77.97983



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71394, 77.99572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71916, 78.01117



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72504, 78.01680



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73766, 78.03445



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03868



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74120, 78.04588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77998, 78.05786



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86915, 78.08382



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87998, 78.08170



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91266, 78.05852



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92653, 78.06517



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93196, 78.06473



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93473, 78.06268



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93735, 78.05344



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95498, 78.04689



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95847, 78.04989



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96011, 78.04944



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96251, 78.05104



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96427, 78.05118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96526, 78.04950



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96637, 78.04923



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96859, 78.05102



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.12010, 78.13282



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.51381, 78.03208



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51395, 78.02897



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.51632, 78.02571



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52154, 78.02425



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52697, 78.00960



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52856, 78.00884



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.55122, 77.97229



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.55312, 77.96538



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.56474, 77.95413



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.57394, 77.89805



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58285, 77.89311



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58407, 77.88826



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.58363, 77.88722



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.58534, 77.88126



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58929, 77.87842



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59188, 77.86980



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59963, 77.86303



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.60200, 77.86284



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.60468, 77.85443



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.60203, 77.85299



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.59337, 77.84801
